Smart Queue Management System (SQMS)

Abstract

In modern public and private service sectors such as banks, hospitals, educational institutions, and government offices, long waiting times and inefficient queue handling are persistent challenges. Conventional manual queue systems often result in overcrowding, mismanagement, lack of transparency, and customer dissatisfaction. To address these issues, this project presents a **Smart Queue Management System (SQMS)** based on embedded system technology.

The proposed system automates the entire queue handling process using a token-based mechanism. Customers receive digital tokens through push-button interaction, and service counters call tokens sequentially using a controlled interface. The system integrates visual indicators such as an LCD display and LED indicators, along with audio alerts using a buzzer to notify customers effectively. An IR sensor is employed to detect customer presence, enabling automatic skipping of absent tokens and ensuring uninterrupted service flow. Additionally, priority service handling is incorporated to manage urgent cases efficiently.

The system is designed using cost-effective, easily available electronic components and demonstrates practical applications of microcontrollers, sensors, actuators, and communication modules. The Smart Queue Management System improves service efficiency, reduces waiting time, ensures fairness, and enhances overall customer experience. Its scalable and modular design makes it suitable for real-world implementation across various service-oriented environments.

Keywords

Smart Queue Management System, Embedded Systems, Arduino Nano, Token-Based System, Automation, IR Sensor, LCD Display, Service Efficiency

1. Introduction

Efficient queue management is a critical challenge in service-based environments. Customers often experience **long waiting times**, crowding, and frustration in banks, hospitals, ticket counters, and government offices. Traditional queues are manually managed, which leads to delays, inefficiency, and lack of monitoring.

Advancements in embedded systems and automation provide a pathway for designing **smart, automated solutions** that can organize queues effectively. The **Smart Queue Management System (SQMS)** is developed to:

- Automate queue handling
- Reduce crowding and waiting time
- Ensure fairness in service delivery

- Improve customer satisfaction

Beyond its practical utility, this project serves as an educational platform for students to **apply theoretical knowledge of microcontrollers, sensors, actuators, and programming** to a tangible problem. It bridges the gap between academic learning and real-world application, emphasizing the importance of technology in daily life.

2. System Overview / Methodology

The Smart Queue Management System (SQMS) is designed using a **modular embedded system architecture**, where the **microcontroller functions as the central processing unit**, coordinating all input, processing, and output operations. The system follows a **token-based methodology**, which provides a structured and transparent approach to queue management. By automating the process of token generation, token calling, and customer notification, the system reduces manual intervention, minimizes errors, and ensures fairness in service delivery.

The overall methodology of SQMS begins with **customer interaction**, where a token button allows an arriving customer to request a service. Upon pressing this button, the microcontroller **generates a unique token number**, which is immediately displayed on a 16×2 LCD, providing the customer with clear feedback about their position in the queue. When a service counter is ready, the staff presses the "Next" button to **call the subsequent token** in sequence. To ensure that the customer is notified, the system activates **audio alerts via a buzzer** and **visual signals through a servo motor and LEDs**, improving response even in crowded or noisy environments.

To further enhance efficiency, the system continuously monitors an **IR sensor** to detect customer presence. If a customer does not respond within a predetermined time, the system **automatically skips the token**, preventing unnecessary delays. Additionally, **priority requests** are handled separately, allowing urgent cases to be served immediately without disrupting the normal queue flow. This **closed-loop methodology** ensures that the queue management process is continuous, reliable, and automated, making it suitable for various service-oriented environments while significantly improving customer satisfaction and operational efficiency.

Overall Methodology

1. Customers request service by pressing a token button.
2. The system generates a unique token number.
3. Token information is displayed on an LCD.
4. Service staff call the next token using a control button.
5. Audio and visual alerts notify the customer.
6. An IR sensor verifies customer presence.
7. Absent customers are automatically skipped.
8. Priority requests are handled separately.

This closed-loop system ensures continuous, smooth, and automated queue management with minimal human intervention.

3. Hardware Components

3.1. Arduino Nano

Overview:

Arduino Nano is a **compact, microcontroller-based development board** that acts as the **brain of the Smart Queue Management System**. It is responsible for executing all instructions, processing inputs from buttons and sensors, and controlling outputs such as displays, buzzers, LEDs, and motors.Its small size makes it ideal for embedded projects where space is limited.

Purpose in the System:

Token Generation: The Arduino Nano receives signals from the token button, generates a unique token number, and updates the display accordingly.

Display Updates: It sends information to the LCD to show the current token number, system status, and priority alerts.

Sensor Management: It reads input from IR sensors to detect whether a customer is present when their token is called.

Motor Control: The board controls servo motors and LED indicators to provide visual cues, ensuring smooth queue management.

Audio Alerts: It triggers the buzzer for audio notifications when a token is called.

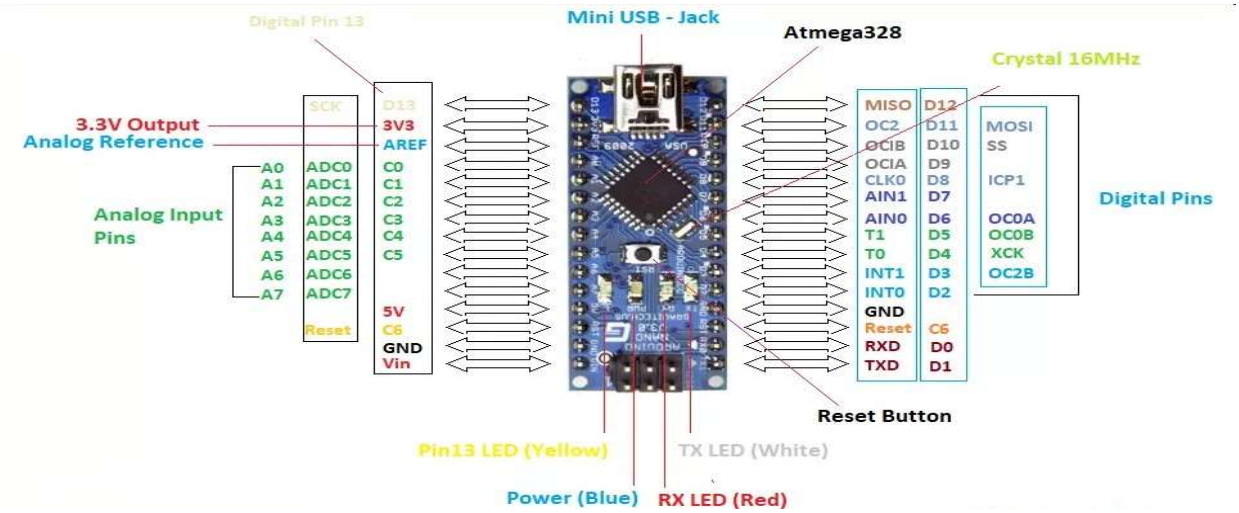

Figure 1: Arduino Nano

Features:

Programmable via Arduino IDE: Easy programming using simple C/C++ language.

Digital and Analog I/O: Can read sensors and control devices simultaneously.

Compatible with Various Modules: Supports LCDs, IR sensors, buttons, LEDs, buzzers, and servo motors.

Summary: Arduino Nano serves as the **central hub** that integrates all hardware components and ensures the Smart Queue Management System works efficiently, reliably, and automatically.

3.2 Push Buttons

Overview:
Push buttons are simple electronic devices that act as a **manual interface** between the user and the system. They are designed to detect a user's physical action, such as pressing or releasing, and convert it into an electrical signal that the microcontroller can read. In the Smart Queue Management System, push buttons allow customers and service staff to interact with the system **without the need for complex input devices** like keyboards or touchscreens.

Figure 2: Push Buttons

Purpose:

Token Generation: When a customer arrives, they press the token button to receive a unique token number, which the system generates automatically.

Calling the Next Token: Service staff press the "Next" button to call the following token in the queue, ensuring organized service.

Priority Service Request: Certain buttons are dedicated for urgent cases, allowing priority customers to skip the normal queue and receive faster service.

Features:

Simple Digital Input: Push buttons provide a straightforward HIGH/LOW digital signal to the microcontroller.

Quick Response: They respond instantly to user input, ensuring smooth operation of the token system.

Easy Integration: Push buttons can be easily connected to Arduino, NodeMCU, or other microcontrollers using basic wiring and coding.

Push buttons are **durable, low-cost, and reliable**, making them ideal for frequent use in public service environments. They also **reduce errors and simplify user interaction**, allowing even untrained users to operate the system efficiently.

3.3 Object Detection IR Sensor

Overview:
The IR (Infrared) sensor is an electronic device that detects the presence or movement of objects, specifically humans, without physical contact. It works by emitting infrared light and detecting its reflection from nearby objects. In the Smart Queue Management System, this sensor plays a crucial role in monitoring customer presence.

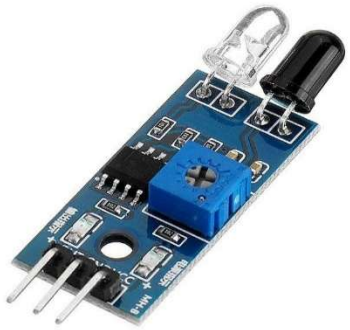

Figure 3: Object Detection IR Sensor

Purpose:

When a service counter calls a token, it is important to know whether the customer is actually present.

The IR sensor automatically **detects the presence or absence** of the person assigned to that token.

This helps the system **decide whether to wait or skip the token**, ensuring smooth and efficient queue management.

Working Principle:

- The sensor continuously emits infrared light.

- If a person is within its detection range, the light reflects back to the sensor.
- The sensor sends a signal to the microcontroller indicating that the customer is present.
- If no reflection is detected within a preset time, the system interprets it as the customer being absent and **moves to the next token automatically**.

Features and Advantages:

- **Non-contact detection:** Can sense humans without physical interaction, keeping the system hygienic and convenient.
- **Fast response time:** Quickly identifies presence, avoiding delays in token processing.
- **Low power consumption:** Operates efficiently without drawing excessive energy, making it suitable for continuous operation.

Significance:

- Ensures **fair and timely service**
- Reduces **waiting time caused by absent customers**
- Enhances the **overall efficiency and automation** of the queue management system

3.4 LCD Display

Overview:
The 16×2 LCD (Liquid Crystal Display) is an essential component in the Smart Queue Management System, providing **visual feedback** to users and service staff. The term "16×2" indicates that the display has **2 rows**, each capable of showing **16 characters**, allowing the system to present clear information in a readable format.

Purpose:
The main purpose of the LCD is to **display real-time information** such as the current token number, service counter status, and system notifications. This ensures that both the customer and the service staff can **easily monitor the progress** of the queue. By providing instant visual feedback, the LCD reduces confusion and improves efficiency.

Figure 4: LCD Display

Features and Explanation:

- **Clear real-time display:**
  The LCD updates the token number instantly whenever a new token is generated or called, ensuring users are **always aware of the current queue status**.
- **Low power usage:**
  LCDs consume very little power, making them **energy-efficient** and suitable for continuous operation in service environments without overloading the system.
- **Supports multiple updates per second:**
  The display can refresh rapidly, which allows the system to **show updated token numbers, skip absent tokens, or display priority notifications** in real-time.

Working in the Project:

- When a customer presses the token button, the LCD shows the **assigned token number**.
- As the service counter calls the next token, the LCD updates to display the **current token being served**.
- Any **priority or skipped token information** can also be displayed, keeping the service flow transparent and organized.

3.5 I2C Module

The **I2C (Inter-Integrated Circuit) module** is a communication interface used to connect multiple devices with just two wires: **SDA (data line)** and **SCL (clock line)**. It is widely used in embedded systems for connecting sensors, displays, and other peripherals with a microcontroller.

Purpose:

In the Smart Queue Management System, the I2C module is primarily used to connect the **LCD display** (or other I2C-compatible devices) to the microcontroller. It simplifies wiring and allows multiple devices to share the same communication lines.

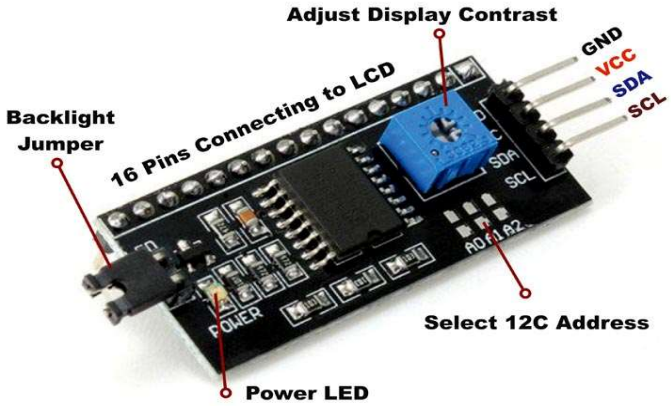

Figure 5: I2C Module

Wiring and Connection:

Connect the **SDA pin** of the I2C module to the microcontroller's SDA pin. Connect the **SCL pin** to the microcontroller's SCL pin. Power the module using **VCC and GND pins**.

Operation:

The microcontroller acts as a **master**, and the connected devices (e.g., LCD) act as **slaves**. Data is transmitted sequentially in bytes, synchronized with the clock (SCL). Each device has a unique **address**, allowing multiple devices to operate on the same bus without conflict.

Advantages in the Project:

Reduces the number of wires compared to traditional parallel LCD connections. Allows easy expansion by connecting more devices like sensors or additional displays. Ensures fast and reliable communication between the microcontroller and peripherals.

3.6 Buzzer

Overview:

- The buzzer is an **audio output device** used in electronic systems to provide sound notifications.
- It is a simple and effective way to **alert users** without requiring them to constantly watch a display.
- In the Smart Queue Management System, the buzzer acts as an **audible signal** whenever a customer's token is called.

Purpose:

- **Notify customers**: When the next token is called, the buzzer sounds to grab the attention of the waiting customer.
- **Enhance system efficiency**: Customers don't have to rely only on visual indicators like LCD displays; they can respond immediately when they hear the alert.
- **Complement visual alerts**: Together with the LCD and LED indicators, the buzzer ensures that all customers are informed effectively.

Figure 6: Buzzer

Features and Advantages:

- **Audible indication**: Produces clear sound, ensuring that the notification is noticeable even in noisy environments.
- **Easy interfacing**: Can be easily connected to microcontrollers like Arduino or NodeMCU, using simple digital output pins.
- **Low-cost**: Buzzers are inexpensive components, making the system budget-friendly.
- **Reliable operation**: They provide consistent alerts over long periods without requiring complex maintenance.
- **Energy-efficient**: Consumes minimal power, making it suitable for continuous operation in service environments.

Step-by-step working in SQMS:

1. The service counter presses the **Next Button**.
2. The microcontroller **sends a signal** to the buzzer.
3. The buzzer **emits a sound** alerting the customer.
4. The customer identifies their token number from the LCD and approaches the counter.

3.7 Servo Motor

Overview:
A servo motor is a type of motor that can rotate to a **specific angle** rather than continuously spinning like a regular motor. In this project, a **180-degree servo motor** is used, which means it can rotate from 0° to 180° precisely, allowing controlled movement in half a circle. This makes it ideal for tasks that require accurate positioning.

Purpose in the System:

Acts as a **visual indicator** to show that a token is being called. For example, it can point to a number or move a small indicator on a display panel. Can be used for **mechanical operations** such as opening or closing a small gate, flap, or barrier when a token is called or skipped. Ensures **interaction between digital signals and mechanical movement**, demonstrating practical embedded system applications.

Figure 5: I2C Module

Features and Advantages:

Accurate angle control: Can move precisely to any angle between 0° and 180°, allowing reliable positioning.

Compact and energy-efficient: Small in size and consumes very little power, suitable for embedded systems.

Reliable performance: Provides consistent movement over repeated cycles, ensuring system accuracy and durability.

Easy to interface with microcontrollers: Can be controlled using simple PWM (Pulse Width Modulation) signals from Arduino or NodeMCU.

Role in the Smart Queue System:

- Provides **real-time feedback** to users.
- Enhances **visual clarity** along with LCD and buzzer alerts.
- Makes the system **interactive and user-friendly**, improving the overall experience.

3.8 LED Indicators

Overview:

LEDs (Light Emitting Diodes) are **small electronic components** that emit light when current passes through them. In the Smart Queue Management System, they serve as **visual indicators** to provide clear, immediate information about the system's status.

Purpose:

LED indicators **communicate the current state of the system** to both customers and staff. These visual cues **reduce confusion** and improve the overall efficiency of queue management.

Features:

LEDs are **bright and highly visible**, ensuring information is easily noticeable from a distance. They have a **long lifespan**, making them reliable for continuous use. LEDs **consume very low power**, which helps maintain energy efficiency and reduces operational cost.

LED indicators are **simple yet essential components** that enhance user awareness and make the system more intuitive and organized.

3.9 SMPS Module

Definition of SMPS:

SMPS stands for **Switched-Mode Power Supply**. It is an electronic power supply that efficiently converts electrical voltage from one level to another using switching regulators instead of linear regulators.

Purpose in this Project:

Most embedded systems, including Arduino, NodeMCU, sensors, and other components, **require a stable 5V DC supply**. The SMPS module converts higher input voltage (like 12V or AC mains) into a **regulated 5V output** suitable for microcontrollers and peripherals.

Working Principle:

Step 1: The input voltage is first rectified and filtered (if AC input) to produce a DC voltage.

Step 2: This DC voltage is then chopped rapidly using a **high-frequency switch.**

Step 3: The chopped voltage passes through an **inductor and capacitor filter**, smoothing it into a stable 5V DC output.

Step 4: A **feedback control mechanism** monitors the output voltage and adjusts switching duty cycle to maintain a constant 5V output even with varying load or input.

Advantages of Using SMPS for 5V Conversion:

- High **efficiency** (~85–95%) compared to linear regulators
- **Compact size** and lightweight
- Provides **stable and regulated voltage** for sensitive electronics
- Reduces **heat generation**

Figure 5: SMPS Module

Application in Smart Queue System:

Powers **Arduino Nano, LCD, IR sensors, LEDs, buzzer, and servo motors** safely and reliably.

4. Software / Algorithm / Code Logic

The software of the **Smart Queue Management System (SQMS)** is developed using the **Arduino IDE**, leveraging embedded **C/C++ programming**. The program acts as the central intelligence of the system, coordinating all hardware components—such as the Arduino Nano, push buttons, IR sensors, LCD, buzzer, servo motor, and LED indicators—to ensure smooth and automated queue management. The algorithm is designed to be **modular, readable, and maintainable**, allowing easy modification, debugging, and expansion for future improvements.

Initialization Phase

Upon powering the system, the software first initializes all connected hardware modules. This includes configuring **digital and analog pins** for input and output, setting up the **I2C communication** for the LCD, and establishing initial states for LEDs, servo motors, and the buzzer. Initialization ensures that all components start in a known, safe state before the system begins operation. For instance, the servo motor is positioned at its default standby angle, the LCD displays a welcome message or "System Ready," and all LEDs are turned off to indicate the system is idle.

Input Handling

The core of the SQMS software is **responsive input detection**. Push buttons serve as the primary interface for both customers and service staff. The program continuously monitors the state of each button in a **loop cycle**, detecting when a button is pressed (HIGH signal). Each button triggers a specific function:

- **Token Button:** Generates a new token for the customer.
- **Next Button:** Calls the next token in the sequence for the service counter.
- **Priority Button:** Adds urgent tokens to the front of the queue for immediate service.

By polling the buttons in real-time, the system ensures that user actions are promptly recognized and processed, avoiding delays and maintaining smooth operation.

Token Generation and Management

When a customer presses the token button, the software automatically generates a **unique token number**. The token counter increments sequentially, storing the token in the internal memory for tracking. Priority tokens are flagged for special handling so they are served ahead of regular tokens. This logic ensures **fairness and order** in the queue, while allowing flexibility for urgent cases.

Output Display and Alerts

Once a token is generated or called, the program updates the **LCD display** with the current token number and relevant system messages. At the same time, it triggers the **buzzer** for audio notification and moves the **servo motor** for mechanical or visual signaling. The LEDs may also be lit to indicate active service or priority tokens. This **multi-modal alert system** ensures that customers are informed in both noisy and crowded environments.

Customer Presence Detection

The software continuously monitors the **IR sensor** at the service counter. When a token is called, the system waits for a preset time for the customer to approach. If the IR sensor does not detect the presence of a customer, the software automatically **skips the absent token** and calls the next one in sequence. This prevents delays and ensures smooth service flow.

Priority Handling

Priority requests are handled through conditional checks in the code. If a priority token exists, it is immediately served before regular tokens. This logic allows the system to manage emergencies or VIP services without disrupting the normal queue order.

Modularity and Readability

The code is organized into **functions** for each task—such as `generateToken()`, `callNextToken()`, `checkIRSensor()`, and `alertCustomer()`—which makes the logic **clear, easy to follow, and maintainable**. Each function is documented with comments explaining its purpose, making the program suitable for demonstration, academic evaluation, and future upgrades.

Pseudocode: Smart Queue Management System

Initialize Hardware and Variables

```
Start
    Initialize LCD
    Initialize Servo
    Set pin modes:
        Token Button -> Input
        Next Button -> Input
        Priority Button -> Input
        Buzzer -> Output
        Status LED -> Output
    Set currentToken = 0
    Set lastIssuedToken = 0
    Set priorityMode = FALSE
    Display "Smart Queue Sys - System Ready" on LCD
    Wait 2 seconds
    Update Display
```

```
End Initialization
```

Main Loop

```
Loop Forever:

    // 1. Generate New Token
    If Token Button is Pressed:
        lastIssuedToken = lastIssuedToken + 1
        Call GenerateTokenEffect()
        Wait 0.3 seconds (debounce)

    // 2. Call Next Customer
    If Next Button is Pressed:
        If currentToken < lastIssuedToken:
            currentToken = currentToken + 1
            Call CallNextEffect()
        Else:
            Display "No More Tokens" on LCD
            Wait 1 second
            Update Display
        Wait 0.3 seconds (debounce)

    // 3. Priority Calling
    If Priority Button is Pressed:
        priorityMode = TRUE
        Display "PRIORITY CALL!" on LCD
        Turn ON Status LED
        Sound Buzzer (1500 Hz, 0.5 sec)
        Wait 1 second
        Turn OFF Status LED
        Update Display
End Loop
```

Subroutine: Update Display

```
Function UpdateDisplay:
    Clear LCD
    Display "Serving: currentToken" on first line
    Display "Wait: lastIssuedToken - currentToken" on second line
    Print "Current Serving: currentToken" on Serial Monitor
End Function
```

Subroutine: Generate Token Effect

```
Function GenerateTokenEffect:
    Turn ON Status LED
    Display "Token Issued!" on LCD
    Display "Your No: lastIssuedToken" on LCD second line
    Sound Buzzer (1000 Hz, 0.1 sec)
    Wait 1 second
    Turn OFF Status LED
    Call UpdateDisplay()
End Function
```

Subroutine: Call Next Effect

```
Function CallNextEffect:
    Sound Buzzer (800 Hz, 0.5 sec)
    Move Servo to 90° (Visual Indicator)
    Call UpdateDisplay()

    Repeat 3 times:
        Turn ON Status LED
        Wait 0.2 sec
        Turn OFF Status LED
        Wait 0.2 sec

    Wait 0.5 sec
    Move Servo back to 0° (Reset)
End Function
```

5. Results and Discussion

The **Smart Queue Management System** was successfully developed and tested in a controlled, simulated service environment to evaluate its performance under typical operational conditions. The testing focused on key functionalities, including token generation, customer calling, priority handling, and audio-visual notification mechanisms.

Observed Results:

- **Smooth Token Generation and Calling:** The system reliably generated sequential tokens for customers and called them in order without errors, ensuring an organized flow.
- **Accurate Detection of Absent Customers:** Customers who did not respond to their token calls were correctly identified, and the system automatically skipped their turn to maintain efficiency.
- **Clear Audio and Visual Alerts:** The integrated buzzer and LCD display provided immediate and unambiguous notification, reducing confusion in the waiting area.
- **Effective Priority Handling:** Urgent or special cases were successfully given precedence, demonstrating the system's flexibility and responsiveness.
- **Stable and Continuous Operation:** Continuous operation over extended testing periods revealed that the system remained stable, with no crashes or malfunctions.

Discussion:

The implementation results indicate that the system **significantly improves the management of queues** compared to traditional manual methods. By automating token distribution and calling, it minimizes human error and reduces overall waiting time for customers. The **automatic skipping feature** ensures that service flow remains uninterrupted, even when some customers are absent. Additionally, the **priority handling capability** allows staff to address urgent cases efficiently, enhancing service quality. The system's **low-cost design and reliable operation** make it highly suitable for deployment in real-world environments such as banks, hospitals, and ticket counters.

Overall, the results validate that the proposed system achieves its primary objective of improving efficiency, organization, and customer satisfaction in service centers.

Code, image and video link:
https://drive.google.com/drive/folders/1GgdgzaOwqG7i1pvo7cQPC20c0zJt8sTX?usp=sharing

6. Conclusion and Future Work

Conclusion

The **Smart Queue Management System** demonstrates a practical and effective approach to improving service delivery in public and private service sectors, such as banks, hospitals, and ticket counters. By automating the token management process, the system minimizes human intervention, reduces waiting times, and enhances overall customer satisfaction.

The integration of **embedded system components**, including microcontrollers, sensors, actuators, and display modules, ensures **reliable and consistent performance**. The project successfully combines **hardware and software solutions** to create a fully functional automated queue management system.

Additionally, the system provides **service transparency** by clearly displaying token numbers and notifying customers promptly, reducing confusion and crowding in service areas. From an academic perspective, this project fulfills its educational objectives by allowing students to gain **hands-on experience with microcontroller programming, interfacing sensors and actuators, and developing a small-scale embedded system**, which strengthens understanding of real-world engineering applications.

Overall, the project highlights the **potential of automation in service management**, combining cost-effectiveness with efficiency and reliability, and serves as a solid foundation for further development in smart service systems.

Future Work

While the current system demonstrates practical efficiency, several improvements can enhance its functionality, scalability, and user-friendliness. Potential future work includes:

- **Integration with Mobile Applications:** Developing a companion mobile app would allow customers to **take tokens remotely, check waiting times, and receive notifications**, further reducing physical waiting and improving convenience.
- **Wireless Communication using IoT:** Connecting multiple counters and service points through **IoT-enabled devices** can provide real-time updates, remote monitoring, and centralized queue management.
- **Database-based Token Storage:** Implementing a **database system** for token tracking would enable historical data analysis, generate reports, and improve system reliability. This could also allow predictive modeling of peak hours.

- **Multi-counter Synchronization:** Enhancing the system to handle **multiple service counters** simultaneously ensures equitable distribution of tokens, reduces bottlenecks, and improves customer flow management.
- **Voice Announcement System:** Incorporating **audio announcements** would assist visually impaired users, enhance accessibility, and improve overall user experience in crowded or noisy environments.

These proposed enhancements indicate the **scalability and adaptability** of the Smart Queue Management System, demonstrating that it can evolve into a more **comprehensive, intelligent, and user-friendly solution** for modern service centers.

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